

Swarat Sarkar

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Summary

Graduate student in Artificial Intelligence with 2+ years of experience developing data-driven solutions and automating processes. Proficient in Python, SQL, cloud platforms, and machine learning frameworks while building analytics pipelines and dashboards.

Education

University at Buffalo | Buffalo, NY, USA

August 2024 – January 2026

Master of Science, Engineering Science – Artificial Intelligence | CGPA: 3.79/4.0

Relevant Coursework: Algorithms Analysis & Design, Intro to ML, Reinforcement Learning, Computer Vision & Image Processing

Indian Institute of Technology Hyderabad | Hyderabad, India

July 2018 – May 2022

B.Tech, Mechanical and Aerospace Engineering | CGPA: 8.1/10 (3.24/4.0)

Relevant Coursework: Data Structures, Calculus, Linear Algebra, Differential Equations, Probability, Statistics, Operations Research

Research & Academic Experience

Research Assistant, UB Safe and Efficient Autonomous Systems Lab | Buffalo, NY, USA

July 2025 – Present

- Driving **Computer Vision pipeline for V2X systems** in **ROS2** using **OpenCV**, **PyTorch**, and **SegNet-based multi-scale CNNs with side-output layers**, enabling real-time RSU–OBU image exchange and feedback via the **Commsignia SDK**.

Software Engineer, UB NASA Lunabotics 2025 | Buffalo, NY, USA

September 2024 – May 2025

- Spearheaded autonomous navigation for **Lunar Rover**, leveraging **ROS2**, **SLAM Toolbox**, **DWBPlanner**, and **Navigation2** on a Raspberry Pi 5. Integrated LiDAR, and motor drivers via Arduino Mega, optimizing real-time localization and path planning.

Professional Experience

Manager IT Network and Communication, TATA STEEL Limited | Jamshedpur, India

April 2023 – August 2024

- Directed a 10+ member team managing **Tata Steel's Network Operations Control** via **OPManager**, monitoring **16,000+ LAN, WAN, and P2P assets**, increasing **uptime to 99.5%** through Wi-Fi analytics and cross-functional issue resolution.
- Designed and implemented dashboards using **Tableau** and **APIs (OPManager, HP Aruba, Cisco Meraki, ServiceNow)** to monitor VIP users and identify recurring incident locations, featured in the **CIO's weekly IT Infra report**.
- Led network setup for a **remote hospital**, implementing **failover uplinks** and **high-availability Wi-Fi** for **5-floor critical care**.
- Analyzed 110k+ incidents using **Python (Pandas, Plotly, NumPy)** with interactive visualizations, reducing outages by 40%.

Management Trainee Systems, TATA STEEL Limited | Remote

August 2022 – April 2023

- Built a responsive web app using **ReactJS**, **ExpressJS**, **Google Maps API**, and **Microsoft SQL** to assist 25k+ users with **real-time bus tracking** and customized maps generated from routes, improving commute in the steel plant.
- Ranked **2nd among 200+ trainees**, earning the Suri Seva Foundation Award, with a **cash prize of 25k rupees**.
- Developed analytics pipelines using **Python**, **R**, deployed cloud ML workflows on **GCP**, **AWS** with **scikit-learn**, **TensorFlow**, and created full-stack apps using **C#**, **Java**, **JavaScript**, **ReactJS**, **.NET**, and databases (**SQL Server**, **Oracle**, **MongoDB**).

Technical Trainee Intern, TATA STEEL Limited | Remote

June 2021 – July 2021

- Crafted UI for the Shop Floor Scheduler using **ReactJS**, **NodeJS**, **ExpressJS**, **MySQL**, incorporating Gantt charts and CRUD operations, reducing manual scheduling interventions by 50%.

Projects

RL-Guided Curriculum Learning Pipeline | *PyTorch, Hugging Face Transformers, Gymnasium, BERT, PPO*

- Devised pipeline for **LLM-based text classification**, using a **PPO agent** to assign sample difficulties, enabling **~60% faster convergence** and improved generalization through **composite reward functions** and BERT embedding analysis.

Real-Time Soccer Offside Detection | *YOLOv8, OpenCV, Computer Vision*

- Developed detection system with **SVD-based** vanishing-point perspective correction and **HSV-based** team classification, achieving **>20 FPS** and **95% accuracy** on consumer GPUs.

Yelp Review Sentiment Analysis | *PySpark, Hadoop, Natural Language Processing*

- Performed **large-scale sentiment analysis** to identify service improvement areas and provide actionable business insights.

Skills

Programming & AI: C/C++, Python, MATLAB, Java, C#, R; Frameworks (TensorFlow, Keras, PyTorch, Gym, XGBoost); Deep Learning (CNNs, RNNs, Autoencoders); Reinforcement Learning (DQN, A3C, PPO); Generative AI (VAEs, Transformers, Diffusion)

Big Data & Data Science: Apache Spark (SQL, MLlib, GraphX), PySpark, Hadoop, Kafka; pandas, NumPy, matplotlib, scikit-learn

Cloud, DevOps & OS: GCP, AWS; Docker, Kubernetes; Jupyter Notebook, VS Code, Git, Postman; Linux (Ubuntu, Raspbian, Kali)

IoT & Computer Vision: Raspberry Pi, Arduino, PLC; OpenCV, ROS2, YOLO; Data Analytics (Python, R); Visualization (Tableau)

Achievements

GRE: 327/340 (Quant: 169, Verbal: 158), **TOEFL:** 105/120 (R: 27, L: 30, S: 24, W: 24)

IIT JEE Advanced 2018: 98.01th percentile (150k+ students) | **IIT JEE Mains 2018:** 98.18th percentile (1.05M+ students)